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Evaluating input data using a deep learning algorithm

Abstract

The invention provides a method for evaluating a set of input data, the input data comprising at least one of: clinical data of a subject; genomic data of a subject; clinical data of a plurality of subjects; and genomic data of a plurality of subjects, using a deep learning algorithm. The method includes obtaining a set of input data, wherein the set of input data comprises raw data arranged into a plurality of data clusters and tuning the deep learning algorithm based on the plurality of data clusters. The deep learning algorithm comprises: an input layer; an output layer; and a plurality of hidden layers. The method further includes performing statistical clustering on the raw data using the deep learning algorithm, thereby generating statistical clusters and obtaining a marker from each statistical cluster. Finally, the set of input data is evaluated based on the markers to derive data of medical relevance in respect of the subject or subjects.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. patent application Ser. No. 16/648,719, filed Mar. 19, 2020, which claims the benefit under 35 U.S.C. § 371 of International Application No. PCT/EP2018/074256 filed on Sep. 10, 2018, which claims the priority benefit of European Application No. EP17192217.2, filed on Sep. 20, 2017, the contents of which are herein incorporated by reference.

FIELD OF THE INVENTION

(1) This invention relates to the field of deep learning, and more specifically to the field of evaluating subject data using deep learning algorithms.

BACKGROUND OF THE INVENTION

(2) Deep Learning is a branch of machine learning that has recently proved to be very successful in the domains of image analysis, speech recognition and natural language processing. Deep learning algorithms model the nonlinear structures in the input data using a sequence of consecutive layers of non-linear transformations. Although they have become popular only recently due to their success in image and speech analysis tasks, they were originally introduced several decades ago in the machine learning literature. These algorithms are mostly successful in tasks where a large number of training labels is available and feature-engineering often requires a large amount of effort from a domain-expert.

(3) In machine learning the quality and number of the training labels can significantly affect the algorithm's performance. Moreover, in many practical applications, the labelling process typically requires substantial effort by domain experts. Thus, there is a natural motivation to optimize this process and label only the data-instances that are critical to optimize the performance of the machine learning algorithm. Active learning focuses on this problem and guides the learning process such that the most informative data-instances are labelled. In the context of deep learning this problem is very challenging due to the complex optimization process that these algorithms employ.

(4) Deep learning methods rely mostly on optimizing certain supervised (classification) loss-functions and, as discussed above, are successful in cases where a large number of training data is available.

(5) There is therefore a need for a means of training a deep learning algorithm to produce consistently accurate results in the absence of a large set of labelled training data. In addition, there is also a need for a means of obtaining user input regarding the progress of deep learning algorithm without requiring extensive effort on the user's part.

SUMMARY OF THE INVENTION

(6) The invention is defined by the claims.

(7) According to examples in accordance with an aspect of the invention, there is provided a computer-implemented method for evaluating a set of input data, the input data comprising at least one of: clinical data of a subject; genomic data of a subject; clinical data of a plurality of subjects; and genomic data of a plurality of subjects, using a deep learning algorithm, the method comprising: obtaining a set of input data, wherein the set of input data comprises raw data arranged into a plurality of data clusters; tuning the deep learning algorithm based on the plurality of data clusters, wherein the deep learning algorithm comprises: an input layer; an output layer; and a plurality of hidden layers; performing statistical clustering on the raw data using the deep learning algorithm, thereby generating statistical clusters; obtaining a marker from each statistical cluster, wherein each marker is a biomarker relating to a single clinical parameter contained within a cluster; and evaluating the set of input data based on the markers to derive data of medical relevance in respect of the subject or subjects, wherein the markers are evaluated with reference to historical subject data collected from subjects with similar conditions and/or symptoms to determine a survival rate and/or an effective treatment method for the subject.

(8) This method evaluates a set of input subject data in order to derive data of medical relevance to the subject using a tuned deep learning algorithm. The deep learning algorithm is tuned based on the existing data clusters of the input data in order to increase the accuracy of the deep learning algorithm. By tuning the algorithm based on the input data, for example, the clinical and genomic data of a subject, the deep learning algorithm may more accurately perform the statistical clustering of the raw data.

(9) Markers are obtained from each statistical cluster and used to evaluate the input data in order to derive data of medical relevance to the subject, for example, a survival rate. The markers may be principal variables of the statistical clusters, used to represent the data contained with said statistical cluster.

(10) The markers may be evaluated with reference to historical subject data collected from subjects with similar conditions and/or symptoms.

(11) In some embodiments, the tuning of the deep learning algorithm comprises of: determining a Gaussian mean width of the input data; determining a convergence rate of a loss-function of the deep learning algorithm; and selecting the number of hidden layers based on the Gaussian mean width and the convergence rate.

(12) In this way, the deep learning algorithm is tuned to account for the manifold structure of the input data in addition to the loss-function of the deep learning algorithm. Thus, optimal number of hidden layers is selected and the accuracy of the deep learning algorithm, and so the evaluation of the input data, is increased.

(13) In further embodiments, the computing of the Gaussian width is based on at least one of the size of the data clusters and the number of data clusters.

(14) In this way, the number of hidden layers of the deep learning algorithm may be selected based on the manifold structure of the input data, thereby further increasing the accuracy of the deep learning algorithm.

(15) In an arrangement, the deep learning algorithm is an autoencoder.

(16) In this way, the deep learning algorithm may perform unsupervised learning on the set of input data in order to learn a representation for the set of input data. By employing an autoencoder, a user is not required to supervise and provide feedback on the learning process of the deep learning algorithm.

(17) In some arrangements, the performing of the statistical clustering comprises considering the hidden layers of the deep learning algorithm as low dimensional representations.

(18) In this way, the hidden layers may be represented using a set of principal variables, thereby reducing the processing power required to operate the deep learning algorithm on the set of input

data.

(19) In an embodiment, the statistical clustering is k-means clustering.

(20) By performing k-means clustering on the raw data, k statistical clusters may be generated to group the raw data points according their position in data space. In this way, it is possible to quickly and efficiently group similar data points.

(21) In an arrangement, the evaluating of the set of input data comprises predicting a survival rate of the subject.

(22) The markers obtained from the statistical clusters may be used to predict the survival rate of the subject by comparing said markers to data collected from previous subjects having similar conditions and/or symptoms.

(23) In an embodiment, the statistical clustering is performed between adjacent hidden layers of the deep learning algorithm.

(24) By using adjacent layers one can identify different structures of clusters that are related to a hierarchical arrangement of data groups. For example, in a cancer class hierarchy, a high level cluster structure can contain two groups one with cancer subjects and the other with the healthy subjects; whereas, more detailed cluster structures may contain each cancer type in a different cluster. It is known that the architecture of deep learning algorithms can capture such hierarchical relations with layers closer to the input data being related to more general structures while deeper layers can capture the more detailed cluster relations.

(25) In an embodiment, the method further comprises: identifying an instance pair in the statistical clusters, wherein the instance pair comprises a pair of raw data points belonging to: different data clusters in the set of input data; and the same statistical cluster after performing the statistical clustering; ranking the instance pair according to an angle between the pair of data points in the input data; presenting the instance pair to a user; obtaining a user input from the user, indicating whether the instance pair should be clustered together; and tuning the deep learning algorithm based on the user input.

(26) An instance pair is a borderline case, where two raw data points were originally clustered into different data clusters when using a closer-to-the-input hidden layer but have been brought into the same statistical cluster when using a deeper hidden layer, for example, by way of k-means clustering.

(27) The angle between the instance pairs, as the raw data may be represented in vector form in the data space, may be used to determine a measure of pairing likelihood. In other words, the greater the angle between two data points, the less likely they will be paired together. Following this, instance pairs having a low angle may be automatically accepted as being correctly clustered together; however, for instance pairs having an angle above a predetermined threshold, a user may be required to confirm that the pairing is correct.

(28) By obtaining input from the user on the borderline pairing cases, and tuning the deep learning algorithm based on said user input, the algorithm may be trained to more accurately cluster and interpret the input data. This in turn leads to a more accurate evaluation of the input data to derive medically relevant information relating to the subject.

(29) In a further embodiment, a threshold to the angle is used to limit the number of instance pairs presented to the user. For example, considering the case with multiple instance pairs that exhibit the previously described behavior (originally clustered into different data clusters when using a closer-to-the-input hidden layer and brought into the same statistical cluster when using a deeper hidden layer) with their angles belonging within the range $[0, \pi]$. A threshold may be set at $2\pi/3$, meaning only the instance pairs that have angles above this threshold are displayed to the user. Alternatively, the threshold may be set at any angle in the range $[0, \pi]$.

(30) In this way, it is possible to ensure that the user is not exposed to an unmanageable amount of data and is only presented with instance pairs having the greatest impact on the tuning of the deep learning algorithm.

(31) According to examples in accordance with an aspect of the invention, there is provided a computer program comprising computer program code means which is adapted, when said computer program is run on a computer, to implement the method described above.

(32) According to examples in accordance with an aspect of the invention, there is provided a controller for controlling the evaluation of a set of input data using a deep learning algorithm, wherein the controller is adapted to: obtain a set of input data, wherein the set of input data comprises raw data arranged into a plurality of data clusters, the input data comprising at least one of: clinical data of a subject; genomic data of a subject; clinical data of a plurality of subjects; and genomic data of a plurality of subjects; tune the deep learning algorithm based on the plurality of data clusters, wherein the deep learning algorithm comprises: an input layer; an output layer; and a plurality of hidden layers; perform statistical clustering on the raw data using the deep learning algorithm, thereby generating statistical clusters; obtain a marker from each statistical cluster, wherein each marker is a biomarker relating to a single clinical parameter contained within a cluster; and evaluate the set of input data based on the markers, wherein the markers are evaluated with reference to historical subject data collected from subjects with similar conditions and/or symptoms to determine a survival rate and/or an effective treatment method for the subject.

(33) In an embodiment, the controller is further adapted to: compute a Gaussian mean width of the input data; compute a convergence rate of a loss-function of the deep learning algorithm; and select the number of hidden layers based on the Gaussian mean width and the convergence rate.

(34) In an arrangement, the controller is further adapted to: identify an instance pair in the statistical clusters, wherein the instance pair comprises a pair of raw data belonging to: different data clusters in the set of input data; and the same statistical cluster after performing the statistical clustering; rank the instance pair according to an angle between the pair of data points in the input data; for an instance pair having an angle greater than a predetermined value, present the instance pair to a user; obtain a user input from the user, indicating whether the instance pair should be clustered together; and tune the deep learning algorithm based on the user input.

(35) According to examples in accordance with an aspect of the invention, there is provided a data analysis system, the system comprising: a storage device, adapted to store the input data; the controller as described above in communication with the storage device; a user interface, adapted to obtain a user input, in communication with the controller; and a display device, adapted to display information to a user, in communication with the controller.

(36) According to yet a further aspect of the invention there is provided a method for evaluating a set of input data, the input data comprising at least one of: clinical data of a subject; genomic data of a subject; clinical data of a plurality of subjects; and genomic data of a plurality of subjects, using a deep learning algorithm, the method comprising: obtaining a set of input data, wherein the set of input data comprises raw data arranged into a plurality of data clusters; tuning the deep learning algorithm based on the plurality of data clusters, wherein the deep learning algorithm comprises: an input layer; an output layer; and a plurality of hidden layers; performing statistical clustering on the raw data using the deep learning algorithm, thereby generating statistical clusters; obtaining a marker from each statistical cluster; and evaluating the set of input data based on the markers to derive data of medical relevance in respect of the subject.

According to yet a further aspect of the invention there is provided a controller for controlling the evaluation of a set of input data using a deep learning algorithm, wherein the controller is adapted to: obtain a set of input data, wherein the set of input data comprises raw data arranged into a plurality of data clusters; tune the deep learning algorithm based on the plurality of data clusters, wherein the deep learning algorithm comprises: an input layer; an output layer; and a plurality of hidden layers; perform statistical clustering on the raw data using the deep learning algorithm, thereby generating statistical clusters; obtain a marker from each statistical cluster; and evaluate the set of input data based on the markers.

(37) These and other aspects of the invention will be apparent from and elucidated with reference to the embodiment(s) described hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Examples of the invention will now be described in detail with reference to the accompanying drawings, in which:
- (2) FIG. 1 shows a method of the invention;
- (3) FIG. 2 shows a visual representation of an autoencoder operating on a set of input data; and
- (4) FIG. 3 shows a data analysis system for executing the method of FIG. 1.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(5) Embodiments of the invention provides a method for evaluating a set of input data, the input data comprising at least one of: clinical data of a subject; genomic data of a subject; clinical data of a plurality of subjects; and genomic data of a plurality of subjects, using a deep learning algorithm. The method includes obtaining a set of input data, wherein the set of input data comprises raw data arranged into a plurality of data clusters and tuning the deep learning algorithm based on the plurality of data clusters. The deep learning algorithm comprises: an input layer; an output layer; and a plurality of hidden layers. The method further includes performing statistical clustering on the raw data using the deep learning algorithm, thereby generating statistical clusters and obtaining a marker from each statistical cluster. Finally, the set of input data is evaluated based on the markers to derive data of medical relevance in respect of the subject or subjects.

(6) FIG. 1 shows a method **100** of evaluating a set of subject input data using a deep learning algorithm.

(7) In step **110**, a set of input data is obtained. The input data comprises raw data arranged into a plurality of data clusters. The raw data may comprise clinical and genomic data of a subject, or plurality of subjects, to be evaluated to derive medically significant data regarding the subject, or subjects.

(8) For example, the data may be evaluated to determine a survival rate of a subject by evaluating the subject's data against a plurality of subjects having the same symptoms or conditions. The raw subject data may, for example, comprise: subject age; blood test results; symptoms; previously diagnosed conditions; and the like.

(9) In step **120**, the deep learning algorithm is tuned based on the plurality of data clusters of the set of input data, wherein the deep learning algorithm comprises: an input layer; an output layer; and a plurality of hidden layers.

(10) The tuning of the deep learning algorithm may comprise: determining a Gaussian mean width of the input data; determining a convergence rate of a loss-function of the deep learning algorithm; and selecting the number of hidden layers based on the Gaussian mean width and the convergence rate. The computing of the Gaussian width may be based on the size of the data clusters and/or the number of data clusters.

(11) Gaussian mean width is formally defined as $E[\sup_{x,y \in K} \langle g, x-y \rangle]$, where g is a standard Gaussian random variable. Intuitively, it provides a measure of size of set K and can be computed for certain frequently used data distribution models, such as: a mixture of Gaussians; sparse signals represented by a dictionary basis; and the like.

(12) In some cases, formulas for a mixture of Gaussians or a sparse dictionary structure may be used to determine several Gaussian mean widths for a given set of input data. The final choice of which Gaussian mean width to use for tuning the deep learning algorithm may be made based on the associated convergence rate for each width.

(13) The deep learning algorithm can be trained with several different architecture choices and the

convergence rate of the loss function can be used as a criterion to select the optimal one, with faster convergence signifying the better architecture.

(14) In step **130**, statistical clustering is performed on the raw data using the deep learning algorithm, thereby generating statistical clusters. The statistical clustering may consider the hidden layers of the deep learning algorithm as low dimensional representations. In other words, principal variables of the hidden layers may be identified to perform the statistical clustering in order to reduce the number of computations required.

(15) The statistical clustering method may, for example, be k-means clustering. K-means clustering is a method of vector quantization, which models a probability density function based on the distribution of vectors, such as the input data. The k-means clustering groups the input data into k different clusters.

(16) Given a set of raw data $(x_{sub.1}, x_{sub.2}, \dots, x_{sub.n})$, where each data point is a d-dimensional real vector, k-means clustering aims to partition the n data points into k, where $k \leq n$, sets $S = \{S_{sub.1}, S_{sub.2}, \dots, S_{sub.k}\}$ so as to minimize the within-cluster sum of squares (WCSS) i.e. the variance. Formally, the objective is to find:

(17)

$$\underset{S}{\operatorname{argmin}} \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2 = \underset{S}{\operatorname{argmin}} \sum_{i=1}^k S_i \cdot \operatorname{Var} S_i,$$

where $\mu_{sub.i}$ is the mean of the data points in set $S_{sub.i}$.

(18) The statistical clustering may be performed between adjacent hidden layers of the deep learning algorithm. By using adjacent layers one can identify different cluster structures that are related to a hierarchical arrangement of data groups. For example, in a cancer class hierarchy, a high level cluster structure can contain two groups one with cancer subjects and the other with the healthy subjects; whereas, more detailed cluster structures may contain each cancer type in a different cluster. It is known that the architecture of deep learning algorithms can capture such hierarchical relations with layers closer to the input data being related to more general structures while deeper layers can capture the more detailed cluster relations.

(19) Following the statistical clustering performed in step **130**, the method may include several additional steps for tuning the deep learning algorithm.

(20) In step **131**, an instance pair is identified within the statistical clusters. An instance pair is defined as a pair of raw data points belonging to different data clusters in the original set of input data and the same statistical cluster after performing the statistical clustering.

(21) In step **133**, the instance pair may be ranked according to an angle between the pair of data points in the input data. The angle can be measured using the vector representation induced by each network layer.

(22) In step **135**, for an instance pair having an angle greater than a threshold angle, the instance pair may be presented to a user. A threshold to the angles can be set to limit the amount of data that a user is exposed to. For example, considering the case with multiple instance pairs that exhibit the previously described behavior (originally clustered into different data clusters when using a closer-to-the-input hidden layer and brought into the same statistical cluster when using a deeper hidden layer) with their angles belonging within the range $[0, \pi]$. A threshold may be set at $2\pi/3$, meaning only the instance pairs that have angles above this threshold are displayed to the user. Alternatively, the threshold may be set at any angle in the range $[0, \pi]$.

(23) In step **137**, a user input may be obtained from the user, indicating whether the instance pair should be clustered together.

(24) For instance pairs that are considered to be unlikely to be paired together, based on their angle in the input data set, the user may provide an input indicating whether the pairing is correct or not. In this way, the deep learning algorithm may also receive supervised input from a trained user in order to increase the accuracy of the algorithm.

(25) In step **139**, the deep learning algorithm may be tuned based on the user input.

(26) By tuning the deep learning algorithm based on the user input, the deep learning algorithm may more accurately assess similar instance pairs in future method cycles, without necessarily requiring additional input from the user. In this way, the deep learning algorithm is made more accurate and efficient.

(27) In step **140**, a marker is obtained from each statistical cluster produced by the deep learning algorithm. As the input data comprises the clinical and/or genomic data of a subject, or plurality of subjects, the markers may be biomarkers relating to a single clinical parameter contained within a given cluster.

(28) The markers may be compared against historic subject records in order to determine which markers have previously served as important subject evaluation points. In this way, the deep learning algorithm may be trained to preferentially identify and evaluate these evaluation points. This is particularly important in cases where only a small amount of data is available for a subject or when the input data is largely unlabeled.

(29) In step **150**, the set of input data is evaluated based on the markers to derive data of medical relevance in respect of the subject or subjects

(30) The markers may be evaluated to derive medically relevant data of a subject, for example, a survival rate of the subject. The survival rate of the subject may be evaluated by comparing the markers to historical data collected from other subjects with similar conditions and/or symptoms. Alternatively, the historic subject data may also include various treatment methods and their effectiveness at treating the known conditions. In this way, the evaluation may also indicate an effective treatment method for the condition of the subject.

(31) FIG. 2 shows a visual representation **200** of an autoencoder **210** performing k-means clustering on a set of input data **220**. It can be seen that the input data **220** comprises raw data points **230** arranged into a plurality of data clusters **240**, represented on a Voronoi diagram.

(32) In this example, the deep learning algorithm is an autoencoder **210**. An autoencoder is an artificial neural network that may be used for unsupervised feature learning. The autoencoder comprises: an input layer **250**; a plurality of hidden layers **260**; and an output layer **270**.

(33) As described above, the hidden layers **260** of the autoencoder **210** are tuned **280** based on the input data. For example, in this case, as there are three data clusters in the original input data, the number of hidden layers of the autoencoder is set to three. Alternatively, the hidden layers may be tuned based on the size of each of the data clusters.

(34) Following the tuning of the autoencoder, the raw data **230** is passed to the input layer **250**. The raw data is then passed through the hidden layers **260** to the output layer **270**. The output layer has the same number of nodes as the input layer and so simply acts to reconstruct the raw data into new statistical clusters **290**.

(35) In this example, the statistical clusters **290** are largely similar to the data clusters **240** of the input data **220**; however, it can be seen that one data point **300** has changed cluster. This data point forms instance pairs **310** with the other data points in the new cluster. These instance pairs may then be ranked according to the angles between the data points in the original data points in the input data set. Those instance pairs above a given threshold angle may be presented to a user in order to obtain user feedback on whether the new clustering of these data points is correct. The autoencoder may then be tuned based on the user's feedback.

(36) FIG. 3 shows an example of a data analysis system **400**. The data analysis system comprises: a storage device **410**, adapted to store the input data; a controller **420** in communication with the storage device; a user interface **430**, adapted to obtain a user input, in communication with the controller; and a display device **440**, adapted to display information to a user, in communication with the controller.

(37) As discussed above, embodiments make use of a controller **420** for performing the data processing steps.

(38) The controller can be implemented in numerous ways, with software and/or hardware, to

perform the various functions required. A processor is one example of a controller which employs one or more microprocessors that may be programmed using software (e.g., microcode) to perform the required functions. A controller may however be implemented with or without employing a processor, and also may be implemented as a combination of dedicated hardware to perform some functions and a processor (e.g., one or more programmed microprocessors and associated circuitry) to perform other functions.

(39) Examples of controller components that may be employed in various embodiments of the present disclosure include, but are not limited to, conventional microprocessors, application specific integrated circuits (ASICs), and field-programmable gate arrays (FPGAs).

(40) In various implementations, a processor or controller may be associated with one or more storage media **410** such as volatile and non-volatile computer memory such as RAM, PROM, EPROM, and EEPROM. The storage media may be encoded with one or more programs that, when executed on one or more processors and/or controllers, perform at the required functions. Various storage media may be fixed within a processor or controller or may be transportable, such that the one or more programs stored thereon can be loaded into a processor or controller.

(41) The user interface **430** may comprise: a mouse; a keyboard; or any other suitable means of obtaining a user input. The display device **440** may comprise a screen.

(42) Other variations to the disclosed embodiments can be understood and effected by those skilled in the art in practicing the claimed invention, from a study of the drawings, the disclosure, and the appended claims. In the claims, the word “comprising” does not exclude other elements or steps, and the indefinite article “a” or “an” does not exclude a plurality. The mere fact that certain measures are recited in mutually different dependent claims does not indicate that a combination of these measures cannot be used to advantage. Any reference signs in the claims should not be construed as limiting the scope.

Claims

1. A computer-implemented method for tuning a deep learning neural network (DLNN) trained using input data comprising clinical data and/or genomic data, the method comprising: performing clustering on the input data using the DLNN to generate a plurality of clusters by identifying principal variables of hidden layers of the DLNN so as to reduce a number of computations required, and considering hidden layers of the DLNN as low dimensional representations; identifying data points that are in different clusters of the plurality of clusters, wherein the identified data points are in: (i) a first hidden layer of the hidden layers of the DLNN and (ii) the same cluster of the plurality of clusters in a second hidden layer of the hidden layers of the DLNN; identifying a borderline case as including the identified data points based on the data points being identified as being in different clusters in the first hidden layer and in the same cluster in the second hidden layer; providing the borderline case to a display; receiving feedback from a user regarding whether the identified data points belong in the same cluster; and updating the DLNN based on the feedback received.
2. The method of claim 1, further comprising: determining a Gaussian mean width of the input data; determining a convergence rate of a loss-function of the DLNN; and selecting hidden layers based on the Gaussian mean width and the convergence rate.
3. The method of claim 2, wherein the determining of the Gaussian mean width is based on the size of the plurality of clusters.
4. The method of claim 2, wherein the determining of the Gaussian mean width is based on the number of the plurality of clusters.
5. A non-transitory computer readable medium having stored thereon a computer program code, wherein such computer program code when run on a computer, causes the computer to execute the method of claim 1.

6. A method as claimed in claim 1, further comprising predicting a survival rate of a subject using the tuned DLNN.
 7. A non-transitory computer readable medium having stored thereon a computer program code, wherein such computer program code when run on a computer, causes the computer to execute the method of claim 6.
 8. The method of claim 1, further comprising identifying a plurality of borderline cases, wherein a threshold to an angle is used to limit the number of borderline cases provided to the user.
 9. The method of claim 1, wherein the clustering includes k-means clustering.
 10. The method of claim 1, wherein the clustering includes clustering between adjacent hidden layers of the DLNN.
 11. The method of claim 1, further comprising: obtaining respective markers from the plurality of clusters, wherein the markers are biomarkers respectively related to clinical parameters respectively contained within the clusters.
 12. The method of claim 11, further comprising: evaluating a set of the input data based on the markers to derive data of medical relevance in respect of a subject or subjects, wherein the markers are evaluated with reference to historical subject data collected from subjects with similar conditions and/or symptoms to determine an effective treatment method.
 13. The method of claim 1, wherein the DLNN is an autoencoder.
 14. A system for tuning a deep learning neural network (DLNN) trained using input data comprising clinical data and/or genomic data, the system comprising: a deep learning neural network (DLNN) trained using input data comprising clinical data and/or genomic data; a display; and a processor configured to: (i) perform clustering on the input data using the DLNN to generate a plurality of clusters by identifying principal variables of hidden layers of the DLNN so as to reduce a number of computations required, and considering hidden layers of the DLNN as low dimensional representations; (ii) identify data points that are in different clusters of the plurality of clusters, wherein the identified data points are in: (1) a first hidden layer of the hidden layers of the DLNN and (2) the same cluster of the plurality of clusters in a second hidden layer of the hidden layers of the DLNN; (iii) identify a borderline case as including the identified data points based on the data points being identified as being in different clusters in the first hidden layer and in the same cluster in the second hidden layer; (iv) provide the borderline case to a display; (v) receive feedback from a user regarding whether the identified data points belong in the same cluster; and (vi) update the DLNN based on the feedback received.
 15. The system of claim 14, wherein the processor is further configured to: determine a Gaussian mean width of the input data; determine a convergence rate of a loss-function of the DLNN; and select hidden layers based on the Gaussian mean width and the convergence rate.
 16. The system of claim 15, wherein determining the Gaussian mean width is based on a size of the plurality of clusters.
 17. The system of claim 15, wherein determining the Gaussian mean width is based on a number of the plurality of clusters.
 18. The system of claim 14, further comprising predicting a survival rate of a subject using the tuned DLNN.
 19. The system of claim 14, wherein the processor is configured to identify a plurality of borderline cases, wherein a threshold to an angle is used to limit the number of borderline cases provided to the user.
 20. The system of claim 14, wherein clustering comprises k-means clustering.
 21. The system of claim 14, wherein clustering comprises clustering between adjacent hidden layers of the DLNN.
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